

Prangan Nandy

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EDUCATION

University of South Carolina

Columbia, SC, USA

Ph.D. in Mechanical Engineering

Jan. 2024 – May 2028 (Expected)

M.S. in Mechanical Engineering, GPA: 4.00/4.00

Aug. 2026

- *Research field: Low-temperature plasma and thermofluid systems*

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh

B.S. in Mechanical Engineering, GPA: 3.60/4.00

Feb. 2017 - May 2022

- *Received Dean's Award in 2021 and 2022*

RESEARCH AND TECHNICAL EXPERIENCE

RASAER Laboratory - University of South Carolina

Columbia, SC, USA

Graduate Research Assistant

Jan. 2024 - Present

- Conducted experimental investigations of non-thermal plasmas (both atmospheric and low-pressure systems) using synchronized ICCD imaging with electrical measurements and optical emission spectroscopy
- Systematized diagnostic techniques for micro discharges generated by piezoelectric transformers
- Built custom reactor configurations and gas handling systems to study plasma in different working gases
- Performed advanced modeling and simulations to investigate plasma discharges, plasma assisted chemical conversion, plasma surface interactions, and plasma plume evolution in non-homogeneous gas mixtures
- Published 2 peer-reviewed journal papers and presented at conferences (GEC and ISPC), effectively communicating research results to academic and professional audiences
- Supervised undergraduate students in laboratory research activities, including literature review, experimental preparation, data processing, and research-presentation development
- Contributed to multi-institution plasma research collaborations with Sandia National Laboratories, the University of Notre Dame, and the University of Michigan by integrating simulations, diagnostics, and mechanistic interpretation across teams

Indetechs Software Limited

Dhaka, Bangladesh

Assistant Engineer

Mar. 2023 - Dec. 2023

- Developed Python-based simulation models for an adiabatic cooling system to estimate outlet temperature and water consumption for geothermal power plant operations
- Standardized Environmental Product Declaration (EPD) and Safety Data Sheet (SDS) templates for material-passport database supporting sustainable product data management

Abul Khair Steel Products Limited

Chittagong, Bangladesh

Technical Management Trainee

Jul. 2022 - Feb. 2023

- Performed root cause analysis of mechanical failures of cold rolling mill which improved system reliability and operational stability
- Designed a roll-changing cart using SolidWorks. After implementation, the new mechanism reduced maintenance time and increased operational profit by \$9000 per week

SKILLS & INTERESTS

- **Simulation & Modeling:** COMSOL Multiphysics, ANSYS/ANSYS Fluent, STAR-CCM+, FactSage, Plasma chemical kinetics
- **Experimental & Diagnostics:** ICCD imaging, OES, Oscilloscope operation, System fabrication, PIV
- **Programming & Data Analysis:** Python, MATLAB, OriginPro, Tecplot 360
- **CAD & Engineering Design:** SolidWorks (Certified SolidWorks Associate - Mechanical Design), AutoCAD
- **Language:** English (Fluent), Bangla (Native), Spanish (Beginner)
- **Interests:** Soccer (Holding midfielder), Birds, Ironman